

OPERA NUMERORUM

NS Tower -- Path A: Phase 103
NS_ESSRescaleNS_PROVED -- Parabolic Scaling Invariance of NS

Author: David Fox | ORCID: 0009-0008-1290-6105 | Date: July 2, 2026

SECTION 1: DOES THIS CLOSE NS (CLAY M6)?

NO. NS_ESSRescaleNS_PROVED closes ONE of 6 remaining deps, reducing to 5. NS_M6_OPEN (Clay prize) requires all 5 remaining deps to be proved. The CRITICAL PATH is NS_CarlemanHeat_OPEN: Carleman estimates for the heat operator via Hormander pseudo-convexity. This is the mathematically deepest step in the ESS 2003 argument and has no shortcut.

SECTION 2: THEOREM -- NS PARABOLIC SCALING INVARIANCE

THEOREM: If $u(x,t)$ solves incompressible NS with pressure $p(x,t)$, then for any $\lambda > 0$:

$u_\lambda(x,t) = \lambda * u(\lambda x, \lambda^2 t)$ $p_\lambda(x,t) = \lambda^2 * p(\lambda x, \lambda^2 t)$ also solves incompressible NS.

SECTION 3: COMPLETE PROOF (6 STEPS, CHAIN RULE)

Step	Calculation	Result
1 Div-free	$\text{div}(u_\lambda)(x,t) = \sum_i \frac{d}{dx_i} [\lambda u_i(\lambda x, \lambda^2 t)]$ $= \lambda * \sum_i \lambda \frac{du_i}{dy_i}(\lambda x, \lambda^2 t)$ $= \lambda^2 * \text{div}(u)(\lambda x, \lambda^2 t)$	$= \lambda^2 * 0 = 0$ OK
2 Time deriv	$\frac{d}{dt} u_\lambda(x,t) = \lambda * \frac{d}{dt} [u(\lambda x, \lambda^2 t)]$ $= \lambda * \lambda^2 * (\frac{d}{dt} u)(\lambda x, \lambda^2 t)$	$= \lambda^3 * \frac{d}{dt} u \text{ at } (\lambda x, \lambda^2 t)$
3 Laplacian	$\Delta u_\lambda(x,t) = \lambda * \sum_i \frac{d^2}{dx_i^2} [\lambda u_i(\lambda x, \lambda^2 t)]$ $= \lambda * \lambda^2 * (\Delta u)(\lambda x, \lambda^2 t)$	$= \lambda^3 * \Delta u \text{ at } (\lambda x, \lambda^2 t)$
4 Nonlinear	$((u_\lambda \cdot \nabla) u_\lambda)_j = \sum_i [\lambda u_i] [\lambda^2 \frac{du_j}{dy_i}]$ $= \lambda^3 * \sum_i u_i (\frac{du_j}{dy_i}) \text{ at } (\lambda x, \lambda^2 t)$	$= \lambda^3 * (u \cdot \nabla) u \text{ at } (\lambda x, \lambda^2 t)$
5 Pressure	$\nabla p_\lambda(x,t) = \nabla [\lambda^2 p(\lambda x, \lambda^2 t)]$ $= \lambda^2 * \lambda * (\nabla p)(\lambda x, \lambda^2 t)$	$= \lambda^3 * \nabla p \text{ at } (\lambda x, \lambda^2 t)$
6 Combine	$[\frac{d}{dt} u_\lambda + (u_\lambda \cdot \nabla) u_\lambda + \nabla p_\lambda - \Delta u_\lambda]_j$ $= \lambda^3 * [\frac{d}{dt} u + (u \cdot \nabla) u + \nabla p - \Delta u]_j (\lambda x, \lambda^2 t)$	$= \lambda^3 * 0 = 0$ QED

SECTION 4: WHY THIS MATTERS FOR ESS 2003

The ESS argument (Escauriaza-Seregin-Sverak 2003) uses parabolic scaling to NORMALIZE a hypothetical blowup scenario. If u blows up at time T^* , one constructs the rescaled sequence $u_{\lambda_k}(x,t) = \lambda_k * u(\lambda_k x, \lambda_k^2 t)$ with $\lambda_k \rightarrow 0$. Each u_{λ_k} solves NS (by this theorem). The sequence converges in $L^{3,\infty}$ to a nonzero 'ancient solution.' Carleman backward uniqueness then forces this ancient solution to be zero -- contradiction. Hence no blowup exists, giving global regularity.

SECTION 5: GAP TABLE -- Phase 103, 5 deps (6 -> 5)

#	Named Open Def	ETA	Critical Path
1	NS_Carleman_SmoothApprox_OPEN	3-6 weeks	NEXT

2	NS_BlowupConcentration_OPEN	2-3 months	ESS centering [+PDE momentum]
3	NS_Carleman_LimitPass_OPEN	2-4 months	Carleman limit
4	NS_CarlemanHeat_OPEN	3-6 months	CRITICAL (Hormander calculus)
5	NS_CarlemanDriftAbsorption_OPEN	after 4	Drift absorption

SECTION 6: CUMULATIVE HISTORY AND INTEGRITY

Phase 95: 7 | Phase 98: 10 | Phase 99: 8 | Phase 100: 8 Phase 101: 7 | Phase 102: 6 | Phase 103: 5 <-- HERE Repository: DavidFox998/navier-stokes New file:
Towers/NS/NSPhase103ESSRescaling.lean (381 lines) Lean toolchain: v4.12.0 | Mathlib: v4.12.0
CMI rules: 0 sorry, 0 axiom keyword, named open defs only M7 manifest: FROZEN (5b80b84d) #print
axioms NS_M6_CLOSED_v103 -> {propext, Classical.choice, Quot.sound}